

When to Use vs Avoid Databases

1. Overview

When designing storage for a system, an important decision is:

None

Use Database ?

or

Use File / Object / Block Storage ?

There is **no universal answer**. Choice depends on:

- Data type
- Query patterns
- Size
- Consistency needs
- Update frequency
- Cost
- Performance
- Scalability
- Operational simplicity

In interviews, always discuss **trade-offs**, not only personal preference.

2. Golden Rule

Use a **database** when you need to **manage data**.

Use **storage systems** when you mainly need to **store and retrieve bytes/files**.

3. Use Database When

A. Structured Data

Examples:

None

Users

Orders

Payments

Inventory

Bookings

Needs schema and relationships.

B. Querying Required

Need filters/search:

SQL

```
SELECT * FROM orders
```

```
WHERE user_id = 101
```

```
AND status = 'PAID';
```

Databases excel here.

C. Transactions Needed

Example:

None

Deduct stock

Create order

Charge payment

Commit

Use SQL DB.

D. Frequent Updates

Rows updated often:

None

balance

cart

status

stock

Databases handle updates better than immutable object stores.

E. Relationships Matter

None

User ↔ Orders ↔ Payments ↔ Coupons

Use relational DB.

4. Avoid Database When

A. Large Binary Files

Examples:

- Images
- Videos
- PDFs
- Audio
- Backups

Better in object/file storage.

B. Massive Static Content

Use:

- Amazon S3
- Google Cloud Storage

C. Streaming Large Files

Databases often load blobs inefficiently.

Better:

None

Store file externally

Stream directly via CDN

D. Very Cheap Cold Storage Needed

Use archive/object storage instead of expensive DB disks.

5. Common Modern Pattern

Use both:

None

Database:

metadata

Object storage:

actual files

Example:

None

users table:

id, name, avatar_url

avatar image:

stored in S3

6. File Size Heuristic (Classic Rule)

Older practical guidance:

Object Size	Suggested Storage
<256 KB	Database often fine
>1 MB	Filesystem/Object storage better
256KB - 1MB	Depends on read/write pattern

Not absolute, but useful interview heuristic.

7. Why Large Files in DB Can Hurt

A. Memory Overhead

Large blobs may be loaded fully into memory.

B. Slow Replication

Leader → replica must copy huge rows.

C. Backup Growth

Database backups become heavy.

D. Slower Queries

Large table pages hurt performance.

8. Example: User Upload Platform

Bad Design

None

```
users table stores 500MB videos
```

Better Design

None

videos table:

id, owner_id, title, s3_url

Actual video stored in Amazon S3

9. Use SQL Database For

- Banking
- Orders
- Billing
- Reservations
- ERP systems
- Inventory

Because correctness > raw storage.

10. Use NoSQL Database For

- Sessions
- Feed data
- User profiles
- Flexible schema
- Massive scale events

Examples:

- MongoDB
- Redis

11. Use Object Storage For

- Images
- Videos
- Logs archive
- Documents

- ML datasets
- Backups

Examples:

- Amazon S3

12. Use Block Storage For

- Database disks
- VM disks
- Low-latency storage volumes

Examples:

- Amazon EBS

13. Decision Matrix

Need	Best Choice
ACID transactions	SQL DB
Flexible JSON docs	Document DB
Fast cache	Redis
Large media files	Object Storage

DB disk volume	Block Storage
Shared mounted files	File Storage

14. Interview Answer Framework

Say:

I'd separate metadata from blob storage. Structured searchable data stays in DB, while large binary assets go to object storage for lower cost and better scalability.

15. Real Production Pattern

None

Frontend



App Server



SQL DB (metadata)

Object Storage (files)

Cache (hot reads)

Search Engine (search)

16. Short Revision Notes

None

Use DB when:

- Query
- Relationships
- Transactions
- Frequent updates

Avoid DB for:

- Huge files
- Static media
- Cheap archive

Use hybrid design.

17. One-Line Summary

Use databases for structured, queryable, transactional data; use file/object/block storage for large binary content, infrastructure storage, and cost-efficient byte storage workloads.